

Unveiling asymmetric carrier dynamics and space-charge compensation in InGaN/GaN multiple quantum wells via capacitance transient spectroscopy

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ABSTRACT

Carrier dynamics, particularly the transition between localized and delocalized states within quantum wells (QWs), are fundamental to the performance of InGaN/GaN optoelectronic devices. In this work, an anomalous capacitance transient phenomenon was observed using a custom-designed photoelectric synergetic transient spectroscopy system. We demonstrate that the thermal activation process of multiple QWs is fundamentally distinct from classical point-defect signatures. Due to polarization-induced asymmetric potential barriers, thermally emitted carriers from 2D ground states to 3D continuum states are not fully released as free charges but remain localized near the space-charge region. These localized carriers act as dynamic compensation centers, triggering a macroscopic redistribution of the space charge to re-establish electrostatic equilibrium. By varying bias conditions, we distinguish the non-equilibrium dynamics of depletion region expansion and contraction. This study provides a new framework for characterizing carrier transport and spatial charge redistribution, offering a definitive clarification of the non-equilibrium physics in nitride-based quantum structures.

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InGaN/GaN multiple quantum well (MQW) p-n junctions are the cornerstone of modern optoelectronic devices, such as high-brightness LEDs, laser diodes, and photodetectors.^{1–3} In the ideal symmetric QW structure, carriers form strictly discrete quantized energy levels along the growth direction. However, phenomena such as subthreshold emission⁴ and efficient escape of photogenerated carriers⁵ in LEDs suggest that the energy levels in QW, which experience band tilting due to the influence of the polarization electric field, may not be strictly discrete, thereby increasing the efficiency of carrier injection and escape from QW.

As devices scale down to the micrometer range (e.g., Micro-LEDs), understanding carrier transport and recombination dynamics under high internal polarization fields has become increasingly critical.^{6,7} Specifically, addressing the asymmetric charge distribution is essential to overcoming the efficiency droop and suppressing non-radiative recombination associated with the size effect.⁸ The trapping,

localization, and escape of carriers associated with deep states have attracted significant interest, as MQW states modulated by strong polarization fields exhibit electronic signatures distinct from those of conventional bulk defects.^{9,10} Establishing this foundational understanding serves as a critical blueprint for overcoming Micro-LED efficiency bottlenecks. By elucidating the physical mechanisms of carrier transport and spatial redistribution in high-field quantum environments, this study provides the essential insights needed to optimize injection efficiency and suppress non-radiative recombination in ultra-scaled nitride devices.

Early interpretations often attributed these signals to nitrogen vacancies (V_N) or related bulk defects.¹¹ However, first-principles calculations and experimental formation energy studies suggest that V_N is unlikely to form spontaneously in n-type GaN.¹² Furthermore, the deep ionization of Mg acceptors in p-GaN frequently leads to carrier freeze out and measurement artifacts at low temperatures.^{13,14}

Previously, anomalous signals, particularly the capacitance transient polarity reversal, were often ascribed to extrinsic factors. For instance, Rigutti *et al.*¹⁵ proposed a model involving piezoelectric-field-driven charge center shifts of point defects, while Schaake *et al.*¹⁶ highlighted the role of donor-like traps at the InGaN/GaN interface. While our recent study¹⁷ successfully identified the intrinsic nature of the 2D–3D state transitions using frequency-domain admittance spectroscopy (AS), the macroscopic electrostatic consequences of these transitions remain to be elucidated. Time-domain transient spectroscopy provides a unique lens to observe the system's return to equilibrium, revealing the kinetic interaction between localized quantum states and the bulk depletion region.

In this work, we extend this physical picture using a custom-designed transient characterization system. The time-domain transient analysis was employed to reveal how the thermally activated carriers trigger an asymmetric space-charge compensation mechanism. We observe anomalous transients with activation energies of 104–119 meV. The slight discrepancy in energy compared to our previous AS results is attributed to the inherent differences between frequency-domain and time-domain measurements, as well as different scanning variables (frequency vs temperature). We demonstrate that these transients emerge from an intrinsic space-charge compensation mechanism governed by polarization-induced barrier asymmetry, providing a complete temporal description of the 2D–3D state interaction.

The epitaxial structure and parameters of the 9-pair InGaN/GaN (2.5/10 nm) MQWs LED sample are shown in Fig. 1(a). The electrical characterization was performed using a custom-built transient spectroscopy system developed on the framework of conventional Deep-Level Transient Spectroscopy (DLTS). Capacitance transients were acquired using an impedance analyzer (MFIA, Zurich Instruments). The bias was pulsed between high and low voltage levels within a range of -2 to 2 V. To ensure the system reached a steady state at each bias level, the high-voltage pulse duration and low-voltage

measurement period were set to 1.2 and 2.0 s, respectively. A small-signal oscillation of 30 mV at 100 kHz was employed for capacitance extraction, with a sampling rate of 53 kHz. The sample temperature was regulated by a Lakeshore Model 336 controller, spanning from 85 to 280 K with a constant ramp rate of 1 K/min.

The energy band structures of 9-pair MQW in Fig. 1(a) at five distinct bias voltages were simulated using the SiLENSe software. The software implements a one-dimensional drift-diffusion model and solves the Schrödinger equation, Poisson equation, and continuity equation by using Dirichlet (or first kind) boundary conditions ($\Psi = 0$) at both boundaries of the computational domain for the Schrödinger equation. The material parameters used in the simulation were all from the default database in the SiLENSe software, with a MQW conduction-to-valence band offset ratio ($\Delta E_C/\Delta E_V$) of approximately 7:3. As shown in Fig. 1(b), it is noteworthy that, in addition to the built-in electric field, significant internal polarization fields within the GaN layers result in a pronounced tilting of the QW energy bands. The electron Fermi level (E_F) intersects the MQW region, while the wells near the p^+ -side remain above E_F due to the high barrier, and the ground states (E_1) of the wells closer to the n -side reside below E_F , characterizing a quasi-neutral state. Figures 1(g) and 1(h) illustrate the protocol for transient capacitance measurements performed by switching between high and low voltages. We define the voltage decrease as the “V decrease” process [Fig. 1(g)] and the voltage increase as the “V increase” process [Fig. 1(h)]. The dashed area in these timing diagrams indicates the specific temporal windows used for transient capacitance data acquisition.

Figure 2 presents the DLTS spectra for four sets of reciprocal V increase (a)–(d) and V decrease (e)–(h) processes. The results reveal a striking symmetry between the two: The V increase processes manifest as positive peaks (rising transients), while the V decrease processes appear as negative valleys (decaying transients) with nearly identical temperature positions and magnitudes. Herein, a positive

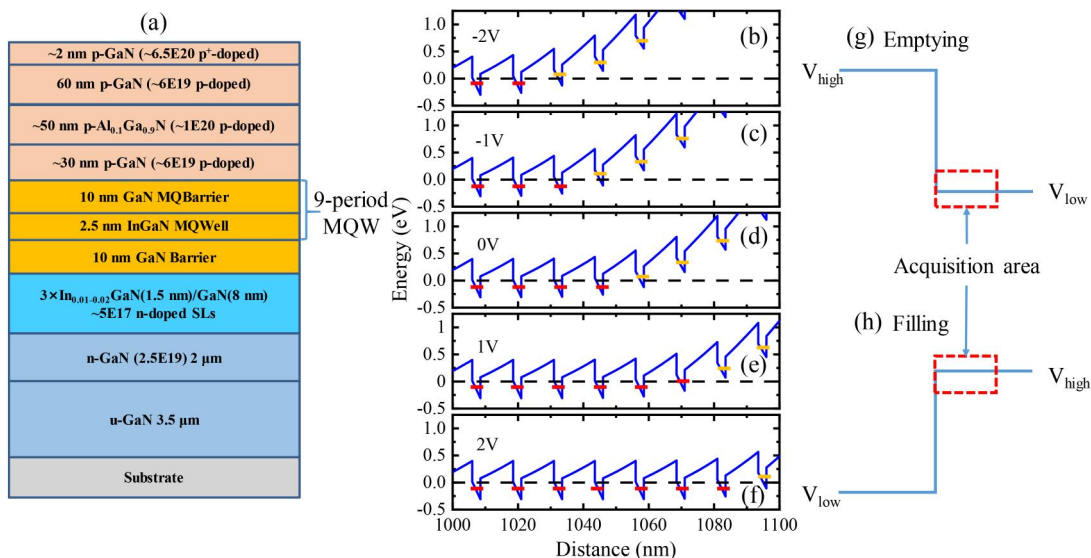


FIG. 1. (a) The epitaxial structure of the InGaN/GaN 9-pair MQWs LED sample. (b)–(f) Simulated energy band diagrams of the p^+ - n junction under five discrete bias voltages. (g) and (h) Timing diagrams illustrating the data acquisition windows for capacitance transients during the “V decrease” and “V increase” processes, respectively.

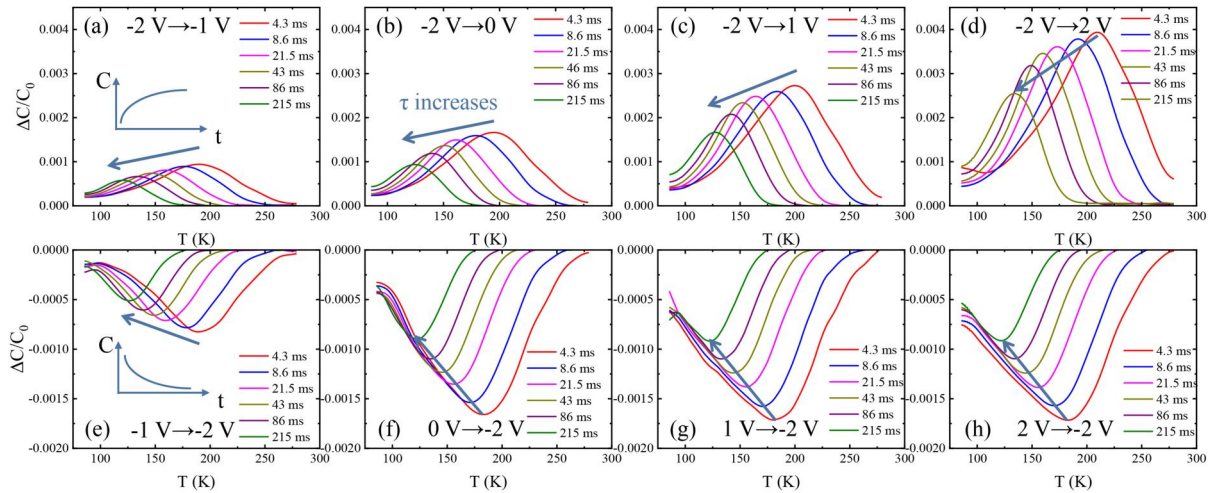


FIG. 2. Capacitance spectra under various time constants for (a)–(d) V increase processes and (e)–(h) the reciprocal V decrease processes. The transient capacitance diagrams in (a) and (e) represent the directions of capacitance change when positive and negative peaks appear, respectively.

peak signifies a rising capacitance transient, whereas a negative valley represents a decaying transient. A detailed quantitative summary of the peak parameters and the timing methodology is provided in Sec. S1 of the [supplementary material](#). As illustrated by the blue arrows, the characteristic peaks exhibit a clear shift toward lower temperatures as the time constant increases.

A closer inspection of the peak morphology reveals subtle bias-dependent variations. When the high voltage is set to zero or negative values [e.g., [Figs. 2\(e\)](#) and [2\(f\)](#)], the peaks are well-resolved and discrete. However, under forward bias [[Figs. 2\(g\)](#) and [2\(h\)](#)], a uniform tail emerges on the low-temperature side of the peaks, which is attributed to a secondary shoulder peak induced by minority carrier injection. Although this tail signal can be isolated through multi-exponential analysis of the transient curves, its intensity remains sufficiently low to avoid significant interference with the primary majority carrier signals; therefore, it is not analyzed in depth in this study.

Most significantly, these observations deviate fundamentally from the classical DLTS signatures characteristic of point defects. A comprehensive comparison between the observed MQW transients and classical point-defect signatures is available in Sec. S2 of the [supplementary material](#) (Table SII). According to conventional theory, the thermal emission of majority carriers (electrons) during the V decrease process typically produces a positive peak. Conversely, our system exhibits anomalous negative peaks for majority carrier dynamics. Furthermore, while carrier capture in point defects (during the V increase process) is generally too rapid to be resolved within standard DLTS rate windows, the MQW system displays positive peaks with intensities and temperature positions nearly identical to those of the negative peaks. This symmetry implies comparable rates for the underlying emission and capture processes. Such pronounced deviations indicate that the observed signals do not originate from discrete trap states—as is the case for point defects—but rather stem from intrinsic charge transport and redistribution mechanisms within the MQW structure, as detailed in the following analysis.

The physical mechanism governing the observed transients is illustrated in [Fig. 3](#). As shown in [Fig. 3\(a\)](#), before being thermally

activated, electrons are strictly confined within the 2D ground states (E_1) of the MQWs. Due to strong polarization-induced fields, a severe barrier asymmetry is created: the n-side barrier height exceeds several hundred meV, whereas the p-side barrier is approximately 100 meV. Consistent with our recent AS study, the 3D continuum states emerge at the well brink, with the continuum edge (E_2) situated above the Fermi level (E_F).

Upon thermal activation [[Fig. 3\(b\)](#)], electrons are excited from the discrete 2D ground states (E_1) into the continuum. It is important to emphasize that E_2 does not represent a discrete, localized energy level; rather, it serves as the demarcation point between 2D and 3D states. Due to the abrupt disparity in the density of states (DOS) between the localized and delocalized regimes, there exists a pronounced kinetic mismatch between the carrier capture and emission processes. Consequently, the thermally activated electrons predominantly reside near the E_2 continuum edge, forming a dynamic charge population that governs the subsequent space-charge redistribution. While these electrons surmount the lower p-side barrier, they remain localized near the edge due to the massive n-side barrier. Unlike discrete point defects, these thermally activated electrons do not immediately escape the depletion region; instead, they act as active space charges that trigger a redistribution of the electrostatic balance.

As illustrated in [Fig. 1](#), there are two QWs located within the space-charge region at a bias of -2 V, with their E_1 remaining below the E_F . For schematic clarity, only one of these wells is depicted in the diagram here. The corresponding thermal activation processes for these QWs are detailed in [Figs. 3\(c\)](#) and [3\(d\)](#). When the bias voltage decreases, the localization of these activated electrons above E_F effectively increases the negative space-charge density within the depletion region. To maintain charge neutrality, the depletion width W must expand to encompass additional ionized donors (ΔN_D^+), leading to the observed negative capacitance transient ($\Delta C < 0$).

Conversely, during the upward voltage step [[Figs. 3\(e\)](#) and [3\(f\)](#)], the compensation of localized electrons near the depletion edge forces the depletion region to narrow more than in the frozen state in order to reach equilibrium, manifesting as a positive capacitance transient

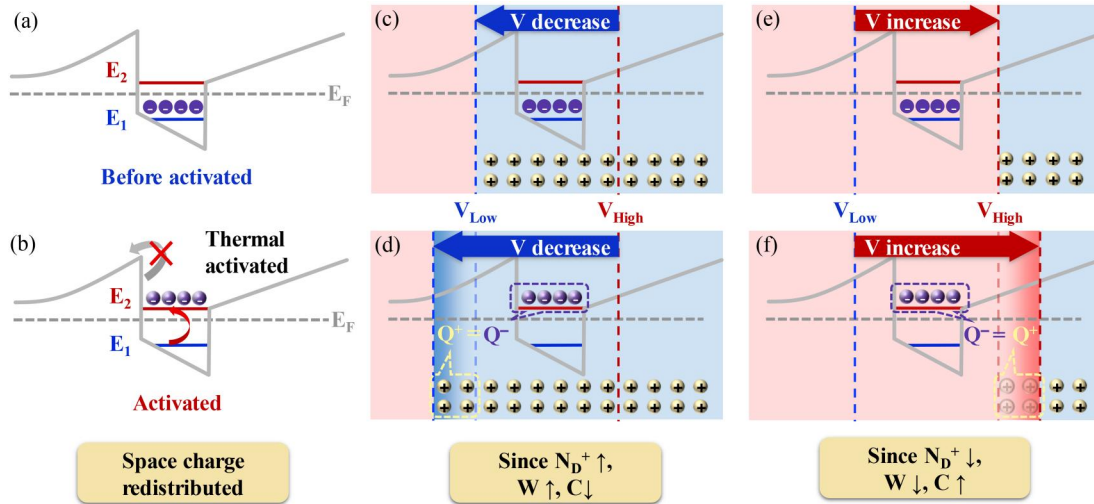


FIG. 3. Schematic of the physical mechanisms governing MQW capacitance transients. (a) Before activated, electrons are “frozen” in the 2D ground states (E_1). (b) Upon thermal activation, electrons transition from 2D states to 3D continuum states (E_2) but remain localized by the n-side barrier. (c) Depletion edge sweep during a V decrease before activated. (d) Activated electrons act as negative space charges, compensating for N_D^+ and forcing depletion expansion, resulting in a negative transient. (e) Depletion edge sweep during a V increase before activated. (f) Compensation of localized electrons leads to depletion contraction and a positive transient.

($\Delta C > 0$). This model successfully links the microscopic 2D–3D transitions to the macroscopic anomalous transients observed in our experiments. To quantitatively evaluate the carrier dynamics within the MQWs, the thermal activation is modeled as the emission of electrons from E_1 to E_2 . The time constant τ is governed by the following analytical expression:¹⁸

$$\tau = \frac{1}{v_c} \left(\frac{2\pi\hbar^2}{m^*kT} \right)^{\frac{1}{2}} \exp\left(\frac{E_2 - E_1}{kT}\right), \quad (1)$$

where m^* is the effective electron mass, k is the Boltzmann constant, and v_c denotes the capture velocity for the E_1 – E_2 transition. Furthermore, the effective charge concentration N_{eff} involved in the transient process is derived from the measured capacitance change ΔC using the depletion approximation

$$N_{\text{eff}} = 2N_D \frac{\Delta C}{C_0}, \quad (2)$$

where N_D represents the background doping concentration and C_0 is the steady-state capacitance.

It is noteworthy that the effective charge concentration N_{eff} reported here is physically distinct from the trap density (N_T) derived in classical DLTS theory. In a conventional DLTS framework, assuming full occupancy, the capacitance transient amplitude corresponds to the total density of states of a discrete trap level. In contrast, for the MQW structures investigated here, the thermal activation process does not involve the entire carrier population within the wells. Instead, governed by occupancy statistics, only a specific subset of carriers is thermally excited to the states that can act as active negative space charges. These “active” charges then trigger the redistribution of the depletion region to maintain electrostatic equilibrium. This statistical limitation explains why N_{eff} is consistently orders of magnitude lower than the total state density.

By fitting the experimental data across various rate windows using these equations, the activation energy and the effective charge concentration are extracted, as summarized in Table I.

Based on the fitting results in Table I, the extracted thermal activation energies ($E_2 - E_1$) for both the V decrease and increase processes are nearly identical, confirming that they originate from the same physical mechanism. The effective charge concentration, N_{eff} , is found to be proportional to the number of participating QWs. As illustrated in Fig. 4(a), different filling voltages correspond to the depletion edge sweeping across a varying number of QWs. By correlating these results with the filling conditions simulated in Fig. 1, the number of involved QWs at specific voltages can be determined (red dots), which exhibits a strong positive correlation with the measured N_{eff} (black dots).

Figure 4(b) illustrates N_{eff} as a function of initial bias following the V decrease process. Despite the variations in the initial pulse voltages, the measurement bias is consistently fixed at -2 V. Under this specific configuration, only two QWs remain positioned below the E_F

TABLE I. Thermally activated energy levels ($E_2 - E_1$), capture velocity (v_c), and effective charge concentration (N_{eff}) at different voltages.

	Bias (V)	$E_2 - E_1$ (meV)	v_c (10^6 cm/s)	N_{eff} (10^{14} cm $^{-3}$)
V $\uparrow$	-2 V $\rightarrow$ -1 V	104 ± 3	3.52	4.35
	-2 V $\rightarrow$ 0 V	110 ± 5	4.35	7.69
	-2 V $\rightarrow$ 1 V	113 ± 2	4.58	12.60
	-2 V $\rightarrow$ 2 V	119 ± 2	4.64	18.12
V $\downarrow$	-1 V $\rightarrow$ -2 V	126 ± 4	14.61	3.80
	0 V $\rightarrow$ -2 V	119 ± 4	11.49	7.65
	1 V $\rightarrow$ -2 V	117 ± 4	10.33	7.92
	2 V $\rightarrow$ -2 V	116 ± 5	9.89	7.89

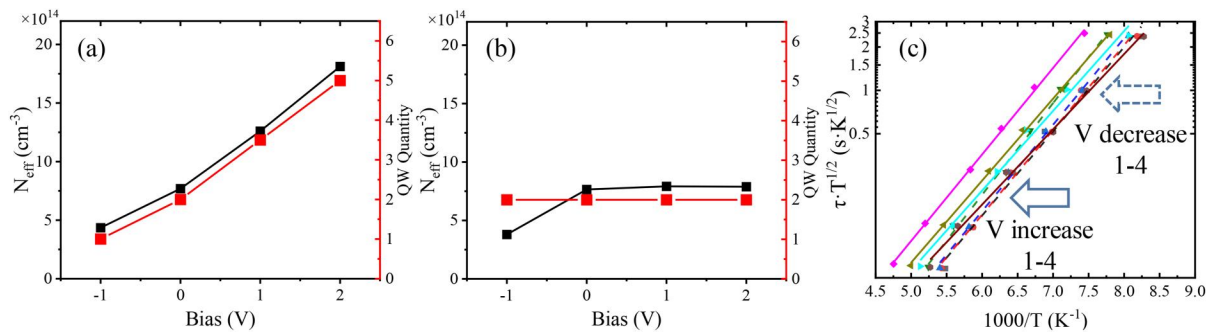


FIG. 4. Effective charge concentration N_{eff} and the corresponding number of involved QWs as a function of bias during the V increase process (a) and V decrease process (b). (c) Arrhenius plots and linear fits for the eight sets of transients, where dashed and solid lines represent the V decrease and increase processes, respectively.

to participate in the thermal emission process. The extracted value of N_{eff} exhibits remarkable stability and does not scale with the pulse magnitude, thereby confirming that only the QWs below the E_F contribute to the observed signal. This characteristic provides a clear distinction between MQW thermal activation and conventional bulk defects; for the latter, all traps located within the entire depletion volume typically participate in the emission, leading to a bias-dependent signal intensity.

Furthermore, the emergence of this thermal activation phenomenon is highly contingent upon the spatial relationship between the MQW region and the depletion edge. If a sample were to contain fewer QWs, or if the applied bias pushed the depletion region entirely beyond the MQW zone, no QWs would remain below E_F to trigger the emission. Consequently, the signature of thermal activation would vanish. This interpretation is consistent with previous literature. For instance, Bourim *et al.*¹⁹ utilized LED samples with only four QWs and observed the thermal activation signal (105 meV) exclusively at a bias of 2 V. In their case, the thermal activation was detectable only when the depletion width was sufficiently narrow to allow the QWs to fall below the E_F . The minor deviation observed at -1 V is likely due to the close proximity of the depletion edges between -1 and -2 V at the boundary of the depletion region.

Finally, Fig. 4(c) summarizes the Arrhenius plots and linear fits for the eight sets of transients. V decrease and increase processes are represented by dashed and solid lines, respectively, with arrows indicating the direction of increasing bias. The consistent slopes confirm the stability across all bias conditions. The slight shifts in the intercepts, which are more pronounced for the V increase process, reflect subtle variations in the v_c . This may be caused by the fact that the V increase process is measured at different voltages, where variations in the local electric field modulate the carrier capture dynamics.

To confirm the observed transients are related to MQW intrinsic states rather than the p-GaN cladding layer, we contrast our results with the known properties of Mg acceptors. The ionization energy of Mg in GaN is typically reported in the range of 150–170 meV,¹⁴ significantly higher than our extracted E_2-E_1 of 104–119 meV. Furthermore, our transients represent a majority-carrier process occurring within the weakly n -type MQW region, as they are triggered without forward-bias injection. This is consistent with our previous AS findings,¹⁷ which are inherently sensitive only to majority-carrier transitions intersecting the Fermi level. Given that Mg levels in the n -region act as minority-carrier traps far from

equilibrium, they are effectively invisible to both AS and the reverse-bias transients reported here. Finally, by analyzing raw capacitance transients (ΔC) rather than normalized signals ($\Delta C/C_0$), we have successfully decoupled the intrinsic MQW dynamics from any parasitic effects arising from p-GaN carrier freeze out.

The physical mechanism of carrier escape in our model is further supported by optical characterizations of similar structures. Marcinkevičius *et al.*²⁰ employed time-resolved photoluminescence to demonstrate that carrier transport in InGaN/GaN MQWs is governed by thermionic emission over barriers modified by polarization fields. While their study focused on the ultra-fast interwell transport on a nanosecond timescale, the underlying thermionic emission framework is entirely consistent with the thermal activation energies extracted from our long-period capacitance transients. In our case, the polarization-induced barrier asymmetry leads to a localization of thermally activated carriers near the space-charge region. These localized carriers act as dynamic compensation centers, modulating the electrostatic equilibrium and giving rise to the anomalous transient polarity, a phenomenon that provides an electrical counterpart to the fast carrier dynamics observed optically.

These insights bridge our physical findings with practical engineering strategies. Specifically, optimizing the polarization-induced barrier heights and tailoring the local doping profiles near the MQW region can mitigate the dynamic space-charge compensation observed here. This provides a clear roadmap for technological applications, allowing for the design of quantum structures that minimize parasitic electrostatic effects and enhance radiative performance.

In conclusion, we have investigated the anomalous capacitance transient behavior in InGaN/GaN MQW p^+-n junctions using a custom-designed transient spectroscopy system. Both the V decrease process and the V increase process exhibit characteristics that deviate fundamentally from classical defect-related carrier emission and capture. Our analysis reveals that when electrons are strictly confined within the QW ground states, their charges are effectively screened, resulting in no observable capacitance transients. However, due to the strain-induced barrier asymmetry, the thermally activated electrons can surmount the lower potential barrier but remain localized near the space-charge region, unable to escape over the higher n -side barrier. These activated electrons, no longer screened, act as dynamic negative charge centers that modulate the electrostatic equilibrium. To re-establish this equilibrium, the depletion width undergoes a temporal evolution, manifesting as the observed capacitance transients.

Quantitative Arrhenius analysis yields consistent activation energies (104–119 meV) corresponding to the QW depth. These findings not only clarify the physical origin of non-ideal thermal activation phenomena in nitride quantum structures but also provide a robust experimental framework for characterizing carrier transport and spatial charge redistribution in QW-based optoelectronic devices.

See the [supplementary material](#) for a detailed quantitative analysis of the capacitance transients in [Fig. 2](#) and a comprehensive comparison between MQW carrier dynamics and classical point-defect signatures.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Zilan Wang and Bo Liu contributed equally to this paper.

Zilan Wang: Conceptualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Bo Liu:** Conceptualization (equal); Formal analysis (equal); Methodology (equal). **Xue-Qi Wang:** Data curation (equal); Formal analysis (equal). **Guochen Wei:** Data curation (equal); Software (equal). **Francis Chi-Chung Ling:** Methodology (equal); Supervision (equal). **Lai Wang:** Funding acquisition (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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Supplementary Material

Unveiling Asymmetric Carrier Dynamics and Space-Charge Compensation in InGaN/GaN Multiple Quantum Wells via Capacitance Transient Spectroscopy

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S1. Quantitative Analysis of Capacitance Transients and Rate-Window Methodology

The capacitance transient data in this work were analyzed using the conventional rate-window (e) methodology. By selecting a series of emission time constants ($\tau = 1/e$), the temperature-dependent shift of the DLTS peaks can be systematically tracked. In our experimental setup, the emission time constant τ is defined using a constant period ratio of $t_2/t_1 = 10$. The transient signal is normalized as $\Delta C/C_0$, where $\Delta C = C(t_2) - C(t_1)$ and C_0 denotes the steady-state capacitance at the corresponding bias.

The thermal activation behavior is characterized via the Arrhenius plot of $\ln(\tau T^{1/2})$ versus $1/T$, as illustrated in Fig. 4(c) of the main text. The activation energy ($E_2 - E_1$) defined in Eq. (1) is extracted directly from the slope of the linear fit. Detailed quantitative parameters, including the peak temperatures (T_{peak}) and transient amplitudes ($\Delta C/C_0$) under various bias conditions, are summarized in Table SI. It is observed that as the emission time constant τ increases (corresponding to a decreasing rate window), the signal peaks systematically shift toward lower temperatures, confirming the thermally activated nature of the 2D-3D carrier transitions.

Table SI. Quantitative information of Figure 2

Bias	-2 V $\rightarrow$ -1 V		-2 V $\rightarrow$ 0 V		-2 V $\rightarrow$ 1 V		- 2 V $\rightarrow$ 2 V	
Time Constant	T_{peak} (K)	$\Delta C/C_0$ ($\times 10^{-4}$)	T_{peak} (K)	$\Delta C/C_0$ ($\times 10^{-4}$)	T_{peak} (K)	$\Delta C/C_0$ ($\times 10^{-4}$)	T_{peak} (K)	$\Delta C/C_0$ ($\times 10^{-4}$)
4.3 ms	190	9.4	195	16.6	200	27.3	210	39.3
8.6 ms	177	8.9	179	15.9	183	25.9	192	37.8
21.5 ms	158	8.1	161	15.0	164	24.8	171	36.0
43 ms	143	7.3	151	13.7	152	23.4	160	34.6
86 ms	134	6.7	139	11.9	140	20.7	148	31.8
215 ms	121	5.8	124	9.4	128	16.7	134	25.5
Bias	-1 V $\rightarrow$ -2 V		0 V $\rightarrow$ -2 V		1 V $\rightarrow$ -2 V		2 V $\rightarrow$ -2 V	
Time Constant	T_{peak} (K)	$\Delta C/C_0$ ($\times 10^{-4}$)	T_{peak} (K)	$\Delta C/C_0$ ($\times 10^{-4}$)	T_{peak} (K)	$\Delta C/C_0$ ($\times 10^{-4}$)	T_{peak} (K)	$\Delta C/C_0$ ($\times 10^{-4}$)
4.3 ms	191	-8.2	185	-16.6	184	-17.1	182	-17.2
8.6 ms	179	-7.8	172	-15.4	170	-15.8	170	-15.7
21.5 ms	158	-7.1	157	-13.5	156	-13.8	155	-13.9

43 ms	150	-6.6	145	-12.3	143	-12.4	145	-12.4
86 ms	141	-6.0	135	-11.0	135	-11.0	134	-11.0
215 ms	129	-5.1	124	-9.0	122	-9.1	121	-9.1

S2. Comparison of Transient Signatures between MQW 2D-3D Dynamics and Classical Point Defects

To further substantiate the origin of the observed signals, Table SII provides a comprehensive comparison between the transient features of InGaN/GaN MQW 2D-3D dynamics and classical point-defect signatures (including both majority and minority traps). The results demonstrate fundamental differences in intensity dependence and physical interpretation, reinforcing that the observed phenomena are intrinsic to the quantum well structure rather than discrete trap levels.

Table SII. Comparison of Capacitance Transient Features

Feature	MQW 2D-3D Dynamics (This Work)	Classical Point Defect (Majority Trap)	Classical Point Defect (Minority Trap)
Transient direction during voltage increase	Increases ($\Delta C > 0$)	Often too fast to resolve	Often too fast to resolve
Transient direction during voltage decrease	Decreases (Negative peak)	Increases (Positive peak)	Decreases (Negative peak)
Bias dependence of signal intensity (N_T)	Proportional to the number of QWs crossing E_F	Proportional to the total depletion volume	Proportional to the total depletion volume
Physical meaning of activation energy	Energy separation from 2D ground state to 3D continuum	Distance from trap level to conduction band ($E_C - E_T$)	Distance from trap level to valence band ($E_T - E_V$)
Linear Arrhenius relationship	$\ln(\tau T^{1/2}) \propto 1/T$	$\ln(\tau T^2) \propto 1/T$	$\ln(\tau T^2) \propto 1/T$